

Danielle Grau Ph.D.

🔗 <https://dgrau13.github.io/>
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EDUCATION

- **Georgia Institute of Technology**
Doctor of Philosophy - Earth & Atmospheric Sciences; GPA: 3.87 Aug 2021 - June 2026
Doctoral Minor: High Performance Computing
- **Florida International University**
Bachelors of Science - Physics; GPA: 3.92 Jan 2018 - May 2021
Honors: Summa Cum Laude, ΦBK

EXPERIENCE

- **Graduate Research Assistant**
Georgia Institute of Technology Aug 2021 - July 2026
 - **Analyze Observational, Remote-Sensing Datasets:** Developed MATLAB and Python-based workflows to analyze the statistical surface roughness properties of the entire Antarctic and Greenland Ice Sheets from NASA ICESat-2 Satellite Altimetry tracks.
 - **Designed and Parallelized Monte-Carlo Simulations:** Developed a MATLAB-based workflow that generates a random, statistically rough surface and uses a cellular automaton model to simulate surface water distribution and routing.
 - **Extrapolate Relationships from Large Simulated Datasets:** Utilized MATLAB's Curve Analysis library to derive mathematical relationships from a Monte-Carlo simulated data set.
 - **Model Configuring and Tuning:** Performed Transient Projections for Antarctic Ice Sheet Loss from 2015 to 2300 with external GCM and Ocean Forcing.
 - **Large Ice-Sheet Model Component Development:** Developed new class and calving scheme within NASA's Ice-Sheet and Sea-Level System Model (ISSM) using MATLAB and C++.
 - **Present Scientific Research:** Present academic research in both poster and oral format at national and international conferences.
 - **Scientific Writing:** Composed and published scientific articles in high-profile journals (*Nature Communications*)
 - **Collaboration:** Collaborated with other professional scientific researchers from other research institutions.
 - **Mentor Undergraduate Researchers:** Provide resources and guidance on research projects for three undergraduate student researchers.
 - **Secured over \$50,000 in fellowships and grants:** Internal fellowships & the NASA Georgia Space Grant Consortium.
- **President, Graduate Students in Earth & Atmospheric Sciences**
Georgia Institute of Technology, School of Earth & Atmospheric Sciences June 2024 - May 2025
 - **Delegation:** Collaborate with other members of the executive board and assign tasks.
 - **Organize Events:** Graduate Student Socials, Professional Development Events, Department-wide events.
 - **Coordinate with Department Faculty and Staff:** Deliver concerns and worries from graduate students and collaborate on solutions.
- **Undergraduate Research Assistant**
Georgia Institute of Technology May 2020 - May 2021
 - **Signal and Power Spectral Density Analysis:** Developed a Python-based workflow to perform PSD analysis on ICESat-2 land-ice Altimetry tracks.
- **Undergraduate Teaching Assistant**
Florida International University Jan 2020 - May 2021
 - **Lecture Aid:** Aid students' comprehension of course materials and host exam review sessions.
 - **Laboratory Aid:** Assist students in laboratory experiments and analysis for Intro. Physics I & II Lab sections.
 - **Exam Proctor:** Assist in proctoring and prepping exam materials.

SKILLS SUMMARY

- **Languages:** Python, MATLAB, Linux, C++, L^AT_EX, Mathematica
- **Frameworks:** Scikit, Pandas, TensorFlow, Keras, PBS, Slurm
- **Tools:** GitHub, Machine Learning, Parallel Computing, Cloud Computing, High Performance Computing, Data Visualization, MS Office, Adobe Illustrator
- **Soft Skills:** Leadership, Event Organization & Management, Scientific & Professional Writing, Public Speaking
- **Linguistics:** English, Spanish, French

PROJECTS

- **NASA Goddard Institute of Space Studies:** Implementing supraglacial melt lake parameterizations into existing albedo scheme for glacier ice in ModelE. (Feb 2024)
- **Patagonian Ice-field Research Program (PIRP):** Measured Calving Events at Grey Glacier, Chile using an Aquarian Audio & Scientific H2d Hydrophone faster. Data was analyzed using signal processing in the field. Tech: MATLAB, Signal Processing. (Nov 2023)
- **ICESat-2 Hackweek 2023:** Develop a Python-based workflow to analyze grounding zone features in Antarctica using NASA's ICESat-2 Land Ice Altimetry. Tech: CryoCloud, SlideRule Earth (Aug 2023)

COMMUNITY ENGAGEMENT & LEADERSHIP EXPERIENCE

- **Graduate Student Senator** Georgia Tech
Graduate Senator for the School of Earth and Atmospheric Sciences Aug 2025 - Present
 - **Review and Vote on Legislation:** Review and Vote on incoming bills from various committees.
- **President of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society May 2020 - Apr 2021
 - **Delegation:** Collaborate with other members of the executive board.
 - **Organize & Host Events:** Professional Development Events, Career Advice, Coding Workshops
- **APS Inclusion, Diversity, and Equity Alliance (Florida International University Member)**
American Physical Society Aug 2020 - Apr 2021
- **Vice President of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society May 2019 - Apr 2020
 - **Coordinate Conference Travel:** Arrange transport, lodging and conference registration for APS March and SEASAPS for physics undergraduate students.
 - **Oversaw Pion Mentorship Program:** Matched senior undergraduate students and junior undergraduate students in mentor-mentees pairings. Organized events for mentors and mentees including professional development, navigating university, and academic advising.
- **Secretary of $\Sigma\Pi\Sigma$ Florida International University**
National Physics and Astronomy Honor Society Sep 2018 - Apr 2019
 - **Record-Keeping:** Kept meeting notes and extensive records of organization meetings and events. Allow kept attendance of all events and meetings.
 - **Communication:** Main point of contact for the organization for both undergraduate students and university departments.
 - **Marketing and Promotion:** Oversaw market and promotions of organizations events and meetings via social media.

PUBLICATIONS

1. **Grau, D.,** M. Poinelli, N.J. Schlegel, H. Seroussi, A.A. Robel. Accelerated Ice Loss from Antarctica in Simulations with Calving Driven by Supraglacial Melt Lake Depth. (Submitted *Geophysical Research Letters*, June 2026).
2. **Grau, D.,** A. Hussain, A.A. Robel (2025). Predicting Mean Depth and Area Fraction of Antarctic Supraglacial Melt Lakes with Physics-Based Parameterizations, *Nature Communications*, In Press.

CONFERENCE PRESENTATIONS

1. "Modeling interaction between supraglacial melt lakes and calving in transient Antarctic simulations", European Geophysical Union Assembly 2026, Oral Presenter. Vienna, Austria. May 2026.
2. "Modeling interaction between supraglacial melt lakes and calving in transient Antarctic simulations", American Geophysical Union Meeting 2025, Oral Presenter. New Orleans, LA, USA. Dec 2025
3. "A Physics-Based Parameterization of Mean Melt Lake Depth and Area Fraction of Supraglacial Melt Lakes", European Geophysical Union Assembly 2025, Poster Presenter. Vienna, Austria, May 2025.
4. "A Physics-Based Parameterization to Predict Mean Depth & Areal Coverage of Supraglacial Melt Ponds", West Antarctic Ice Sheet Work Shop, Poster Presenter. Cloquet, MN, USA. Sep 2023.
5. "A Physics-Based Parameterization to Predict Mean Depth and Areal Coverage of Supraglacial Melt Ponds", Climate Sustainability Challenges and Opportunities Workshop, Oral Presenter. Atlanta, GA, USA. Aug 2023.
6. "A Statistical Parameterization of Supraglacial Melt Pond Are and Depth on Fractal Ice Sheet Surfaces from Percolation Theory (1160496)", American Geophysical Union Annual Meeting 2022, Poster Presenter. Chicago, IL, USA. Dec 2022.
7. "Statistical Models of Supraglacial Melt Ponds Characteristics from Monte-Carlo Simulations", Women in High-Performance Computing, Poster Presenter. Atlanta, GA, USA. April 2022.